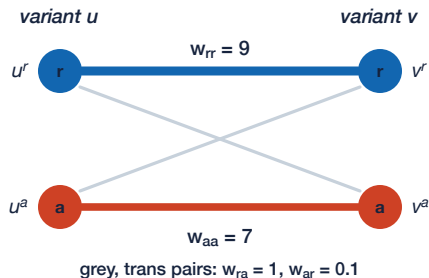


a Allele-pair weights between two variants



$$P = w_{rr} + w_{aa} = 16 \quad \text{cis}$$

$$Q = w_{ra} + w_{ar} = 1.1 \quad \text{trans}$$

$$s = \min(P, Q) / \max(P, Q) = 0.07$$

each read adds 1 (both bases $\geq Q12$) or 0.1

d Exported graph (DOT)

```
1001.1 -> 1543.1 [label=16.0]
1001.2 -> 1543.2 [label=1.1]
```

position.allele → position.allele, label = vote weight

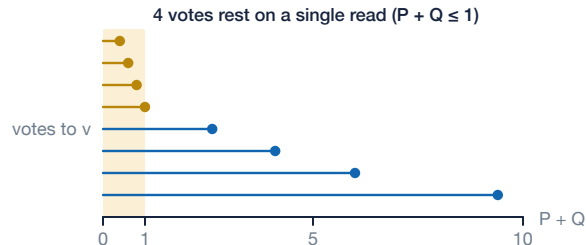
b Vote rules for the pair (*u*, *v*)

condition on <i>s</i>	vote for <i>v</i>
$s > 0.7$	no vote
$s > 0.3$, SNV-5mC pair	no vote
$0.1 < s \leq 0.7$	vote larger of <i>P</i> , <i>Q</i> ; weight 1
$s \leq 0.1$ or one side unsupported	vote; weight 20
<i>u</i> is a tandem-repeat indel	vote; weight 0.1

the vote names a haplotype for *v* given hap(*u*)

$P \rightarrow v^r$ on hap(u^r); $Q \rightarrow v^r$ on hap(u^a)

c Single-read guard



> 3 such votes → recompute h_1 , h_2 from $s < 0.2$, non-indel votes only